

1. The joint probability density function (결합확률밀도함수) of random variables X and Y is given as

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 0, & \text{elsewhere;} \end{cases}$$

Answer the following questions.

- (1) Marginal probability density function (여백밀도함수) of X is given as

$$f(x) = \begin{cases} ax + b, & 0 \leq x \leq 1; \\ 0, & \text{elsewhere;} \end{cases}$$

Find a and b .

- (2) Calculate $E(X)$ and $Var(X)$.

- (3) Conditioned on $X = x$, calculate the conditional probability density function (확률 밀도함수) $f(y|x)$.

$$1) f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 x + y dy = [xy + \frac{1}{2}y^2]_0^1 = x + \frac{1}{2} = ax + b \quad \therefore a=1, b=\frac{1}{2}$$

$$2) E(X) = \int_0^1 x(x + \frac{1}{2}) dx = [\frac{1}{3}x^3 + \frac{1}{4}x^2]_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$Var(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^1 x^2(x + \frac{1}{2}) dx = [\frac{1}{4}x^4 + \frac{1}{6}x^3]_0^1 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$3) f(y|x) = \frac{f(x, y)}{f(x)} = \frac{x + y}{x + \frac{1}{2}} = \frac{2x + 2y}{2x + 1} \quad \therefore f(y|x) = \int_0^{\frac{2x+2y}{2x+1}} \frac{2x+2y}{2x+1} \cdot 0 \leq y \leq 1$$

2. Assume that the word 'free' occurs in 70% of the spam e-mail. We also assume that the word 'free' occurs in 20% of the non-spam e-mails. It is known that 10% of the received e-mails are considered as a spam e-mail. Answer the following questions.

- (1) What is a probability that the randomly chosen email has a word 'free'?
- (2) Assuming that a new e-mail containing a word 'free', what is the probability that it is spam e-mail ?

spam	spam ^c	'free' spam	'free' spam ^c
0.1	0.9	0.7	0.2

$$1) p('free') = p(spam) \times p('free'|spam) + p(spam^c) \times p('free'|spam^c)$$

$$= 0.1 \times 0.7 + 0.9 \times 0.2 = 0.07 + 0.18 = 0.25$$

$$2) p(spam|'free') = \frac{p(spam, 'free')}{p('free')} = \frac{0.1 \times 0.7}{0.25} = \frac{0.07}{0.25} = 0.28$$

3. Consider an Exponential random variable X of mean $1/3$.

- (1) Calculate the probability density function of X at $X = x$.

- (2) Calculate the moment generating function $M_X(t) = E(e^{tX})$ for $3 - t > 0$.

- (3) Using $M_X(t)$ calculated in the above, calculate $E(X^2)$.

$$1) X \sim \exp(3) : f(x) = 3 \cdot e^{-3x}, x \geq 0$$

$$2) M_X(t) = E(e^{tX}) = \int_0^{\infty} e^{tx} \cdot 3e^{-3x} dx = 3 \int_0^{\infty} e^{-(3-t)x} dx = 3 \left[-\frac{1}{3-t} e^{-(3-t)x} \right]_0^{\infty} = 3 \left(0 + \frac{1}{3-t} \right) = \frac{3}{3-t} = 3(3-t)^{-1}$$

$$3) E(X^2) = M_X''(0) = 6 \cdot 3^{-3} = \frac{6}{27} = \frac{2}{9}$$

$$M_X'(t) = 3 \cdot (-1) \cdot (-1) \cdot (3-t)^{-2} = 3(3-t)^{-2}$$

$$M_X''(t) = 3 \cdot (-2) \cdot (-1) \cdot (3-t)^{-3} = 6(3-t)^{-3}$$

2) Let $X_1, \dots, X_n$ be a random sample from the **exponential** distribution with the p.d.f.

$$f(x|\theta) = \theta e^{-\theta x}, \quad x > 0, \quad \theta > 0.$$

a) Find the **likelihood function** $L(\theta|x_1, \dots, x_n)$. (hint : $L(\theta|x_1, \dots, x_n) = f(x_1|\theta) \cdots f(x_n|\theta)$)

$$a) \quad L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n \cdot e^{-\theta \sum_{i=1}^n x_i}$$

b) Find the **MLE (maximum likelihood estimator)** $\hat{\theta}_n$ of θ .

$$\text{(hint : } \partial \ln L(\theta|x_1, \dots, x_n) / \partial \theta = \sum_{i=1}^n \partial [\ln f(x_i|\theta)] / \partial \theta = 0 \text{)}$$

$$b) \quad \ln L(\theta|x_1, \dots, x_n) = \ln \theta^n \cdot e^{-\theta \sum_{i=1}^n x_i} = n \ln \theta - \theta \sum_{i=1}^n x_i$$

$$\frac{\partial}{\partial \theta} n \ln \theta - \theta \sum_{i=1}^n x_i = \frac{1}{\theta} \cdot n - \sum_{i=1}^n x_i = 0 \quad \sum_{i=1}^n x_i = \frac{n}{\theta} \rightarrow \hat{\theta}_{MLE} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

c) Find the approximate **95% confidence interval** for θ for large n .

(hint: $\hat{\theta}_n \sim N(\theta, [nI_1(\theta)]^{-1})$ for large n ; i.e. $P(-1.96 < Z_n < 1.96) = .95$, $Z_n = [\hat{\theta}_n - \theta] / [1/n I_1(\hat{\theta})]^{1/2} \sim N(0,1)$

where $I_1(\theta) = E[-\partial^2 \ln f(X_1|\theta) / \partial \theta^2 | \theta]$, $E(X_1|\theta) = 1/\theta$, $Var(X_1|\theta) = 1/\theta^2$)

$$c) \quad n \uparrow \quad \hat{\theta}_n \sim N(\theta, \frac{1}{n I_1(\theta)})$$

$$P(-1.96 < Z_n < 1.96) = 0.95 \quad Z_n = \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{1}{n I_1(\theta)}}} \sim N(0,1)$$

$$I_1(\theta) = E\left(-\frac{\partial^2 \ln f(X_1|\theta)}{\partial \theta^2}\right) \quad E(X_1|\theta) = \frac{1}{\theta}, \quad Var(X_1|\theta) = \frac{1}{\theta^2}$$

$$\textcircled{1} \quad I_1(\theta) = E\left(-\frac{\partial^2 \ln f(X_1|\theta)}{\partial \theta^2}\right) = E\left(\frac{1}{\theta^2}\right) = \frac{1}{\theta^2}$$

$$\textcircled{2} \quad \hat{\theta}_n \sim N(\theta, \frac{1}{n I_1(\theta)}) = N(\theta, \frac{\theta^2}{n})$$

$$\textcircled{3} \quad Z_n = \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{1}{n I_1(\theta)}}} = \frac{\hat{\theta}_n - \theta}{\hat{\theta}_n / \sqrt{n}} \sim N(0,1)$$

$$\textcircled{4} \quad P(-1.96 < \frac{\hat{\theta}_n - \theta}{\hat{\theta}_n / \sqrt{n}} < 1.96) = P(\hat{\theta}_n - 1.96 \times \frac{\hat{\theta}_n}{\sqrt{n}} < \theta < \hat{\theta}_n + 1.96 \times \frac{\hat{\theta}_n}{\sqrt{n}}) \quad \therefore \text{CI: } \hat{\theta}_n \pm 1.96 \cdot \frac{\hat{\theta}_n}{\sqrt{n}}$$